

Peter Asanov

SENIOR SOFTWARE ENGINEER

Amsterdam, Netherlands

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About

Senior software engineer with 20+ years of experience, including over a decade building real-time, data-intensive, and mission-critical distributed systems from architecture through implementation.

Skills

Distributed Systems HA/fault tolerance, scalability, partitioning/replication, EDA and messaging, consensus (Raft, Paxos), CAP trade-offs/consistency models

Development

C++11-23 Boost/Asio, Cap'n Proto, FlatBuffers, Seastar

Rust Tokio, hyper, PyO3/maturin

Go Gin, zap/slog

Python FastAPI/Pydantic, SQLAlchemy, Celery, NumPy, pandas, Polars

JVM Java/Scala/Kotlin, Spring Boot/Hibernate, Akka

JS/TS React, Vue, Vite/Vitest, tRPC, Prisma

Build Nix, CMake/Ninja, Conan, Maven, Bazel, Meson

Computer Systems

CPU/concurrency x86-64/ARM64 assembly, SIMD (SSE/NEON), memory models, memory and instruction ordering, fences/barriers, multithreading/multiprocessing HW-aware optimization

Linux task scheduling, VM/paging, seccomp/SELinux, io_uring

Networking TCP/IP, UDP, DNS, routing/NAT, QoS, L4/L7 load balancing, virtual/overlay networks (Tailscale), eBPF/XDP/AF_XDP, Unix/raw sockets

Platform & Infrastructure

Cloud AWS (EC2/ECS/Lambda), Azure (VM/Functions), GCP (GCE/App Engine/Functions)

Virtualization/containers Proxmox, QEMU/KVM, Cloud Hypervisor, LXC/Incus, Docker Engine/containerd

Bare-metal/edge iPXE, Matchbox (Terraform provider), Flatcar/systemd-sysext, Keylime (TPM 2.0 attestation)

Kubernetes Talos Linux, k3s, EKS, AKS, GKE, OpenShift, Helm, Kustomize Gateway API, Cilium (CNI), kube-vip, MetalLB Strimzi, CloudNativePG, cert-manager, Trivy Operator, Prometheus Operator, kube-state-metrics

IaC Terraform, Tofu Controller, Ansible, CDK, CloudFormation, Bicep/ARM templates, NixOps4

CI/CD & GitOps Flux CD, Argo CD, GitHub Actions, GitLab CI/CD, Azure DevOps

Observability Prometheus, Grafana/LGTM, OTel/OTLP, CloudWatch/X-Ray, Azure Monitor/App Insights

IAM/IdPs OAuth 2.0/OIDC (PKCE/mTLS/DPoP), SAML, Entra ID/Azure AD, Cognito, Auth0

Access RBAC, PoLP, tenant isolation, Zero Trust

Data	
DBs	Postgres (PostGIS, TimescaleDB), DynamoDB, MySQL, Cassandra, MongoDB, DuckDB
KV/cache	Redis, etcd, RocksDB, Valkey
Object storage	S3, MinIO, RustFS, Azure Blob, GCS
Messaging	Kafka, RabbitMQ, Pulsar
Pipelines	Airflow, Spark/PySpark, Flink, NiFi
AI/ML	
NLP	POS tagging, lemmatization, NER, NLTK, spaCy, Stanza
LLM integration	RAG, retrieval-ready data, agent orchestration, stateful multi-turn workflows
Cryptography	Curve25519, ECDH, EdDSA, Ristretto, ZK-SNARKs/ZK-STARKs

Work Experience

Senior Software Engineer

Rotterdam, Netherlands

THE OCEAN CLEANUP

Mar 2026 – Jul 2026

Owned infrastructure for the Onboard Compute platform, an edge system for UAV and vessel data ingestion, visualization, operational monitoring, and offshore cleanup planning.

- Scaled k3s from one node to a three-server HA cluster with an embedded etcd quorum; added Kubernetes API endpoint failover and an ingress VIP reachable from both the onboard LAN and tailnet.
- Introduced a two-layer automation model: GitOps for in-cluster state; Ansible for node and access provisioning and k3s/Flux bootstrap. Secured runtime secrets with SOPS and enforced validation across both repositories, including KubeLinter checks of fully rendered Flux trees and kubeconform validation in strict mode against pinned schemas.
- Authored an internal RFC to standardize Gateway API routing across company systems and developed a Talos Linux PoC for immutable, API-managed edge nodes.
- Documented cluster setup and steady-state operations, and wrote recovery runbooks.
- Instrumented UAV ingestion for Zenoh freshness and forwarding, Redis Streams/DLQ processing, and end-to-end latency; added alerts for Flux reconciliation, ingress health, and remote-write failures.
- Built a topology-equivalent local k3s environment with Tart VMs to validate changes without access to the field cluster.

Achievements:

- Enabled unmodified amd64 workloads on Apple Silicon by replacing k3s's embedded containerd with Docker Engine and registering Rosetta through binfmt_misc, bypassing an OCI runtime/seccomp incompatibility.
- Improved frontend map legibility by refining higher-contrast density bands and the visual hierarchy between vessel and route layers.

Platform & services: k3s, Flux CD, kube-vip, MetalLB (L2), Tailscale, Traefik, CloudNativePG, PostgreSQL (PostGIS, TimescaleDB), Redis Streams, Zenoh, Prometheus, Grafana, RustFS

Infrastructure tooling: Ansible, Kustomize, SOPS, KubeLinter, kubeconform, Docker Engine, Tart, Rosetta, binfmt_misc, Nix/flake-parts

App stack: Python, TypeScript, Vue

Senior Software Engineer | Partner Integrations

Amsterdam, Netherlands

Vox AI

July 2025 – Oct 2025

Built backend integrations for a voice-AI drive-thru platform, working directly with founders under two-week delivery windows.

1. Streamorders.com DSP integration: OAuth 2.0 server and location-management API

Implemented serverless AWS infrastructure for the partner data-exchange protocol, including authentication, tracing, and deployment automation.

- Built an AWS Cognito-based OAuth 2.0/OIDC service with PKCE enforcement and compatibility handling for non-compliant partner clients.
- Added correlation IDs for end-to-end tracing across authorization, token, and webhook flows.

- Built a multi-tenant location-management API using a DynamoDB repository pattern and AWS X-Ray integration.
- Structured the CDK stack to avoid 15–40-minute Cognito/CloudFront reprovisioning during iteration.
- Added X-Ray distributed tracing, correlation IDs in logs and headers, AWS Lambda Powertools, and EventBridge/SQS monitoring for OAuth events.

2. POS menu transformation system for Redcat integration

Built a deterministic pipeline that converted flat POS data into nested and reference structures for voice-ordering RAG. Resolved ambiguity, circular dependencies, and orphaned references with transformations resilient to upstream schema changes.

3. RAG data-quality engineering and CI integration

Converted noisy POS inputs into retrieval-ready structures and added CI validation to catch regressions from schema and data changes.

Achievements:

- Delivered a working OAuth 2.0 integration bridging AWS Cognito and a third-party DSP protocol within two weeks.
- Built and tested a RAG-oriented menu mapper that handled poor data quality and complex hierarchical and reference relationships.
- Implemented correlation ID tracing that provided sub-second visibility across distributed OAuth flows.
- Integrated RAG data-quality checks into CI workflows to catch regressions from upstream POS schema and data changes.

Platform & services: AWS Lambda, API Gateway, Cognito, DynamoDB, X-Ray, CloudWatch, EventBridge, SQS, CDK (IaC)

App stack: TypeScript, Python 3.12+, Pydantic, FastAPI, Mangum, Moto, pytest/pytest-asyncio, Vitest, HTTPX, uv

Senior Software Engineer

Netherlands

FREELANCE / CONTRACTING

2022 – Present

Delivered platform, data, and systems engineering across client and open-source projects:

- Open-source contribution (tailscale/tailscale): landed Unix socket backend support for `tailscale serve unix:/path/to/socket` across `cmd/tailscale` and `ipn`, enabling services on local sockets to be exposed over Tailscale without TCP bridging and with safeguards for sensitive sockets.
- Open-source contribution (blacktop/ipsw): fixed demangling of internal symbols, including `.cold.*`, to improve the readability and correctness of disassembly output.
- Agentic protocol contribution (opencode-antigravity-auth): reverse-engineered signed payload handling for Claude 4.6 interleaved thinking in a private Google Antigravity API passthrough; preserved valid signatures and removed only invalid ones to prevent model instability and reasoning-state leakage.
- Deliverect, full-cycle restaurant automation PaaS: extended the ordering data model and API behavior.
Stack: Python, Eve, Celery, Kubernetes (GKE), TypeScript, React, Redis, MinIO (local backend), Terraform, Helm
- Ahold Delhaize (Albert Heijn): online data-ingestion platform for internal compliance and R&D analytics.
Stack: Go, Kubernetes (AKS), Strimzi (Kafka), GitOps (Argo CD), Airflow, Spark, Terraform, Helm
- Private customer: taxi-terminal telemetry ingestion for compliance and fraud analysis.
Stack: Go, Kubernetes (AKS), Kafka, PostgreSQL, Terraform, Helm
- LEGO: internal IAM and identity-provider functionality.
Stack: Azure Functions, Azure SAML identity provider, Python
- Private customer: cloud-based video-conversion service.
Stack: C++20, CMake, Ninja, Go, gRPC-Gateway, FFmpeg, Kubernetes (EKS), CloudWatch, Airflow, Nix, Terraform (Terranix), Helm
- Private customer: custom video-calling service.
Stack: C++23, CMake, Ninja, Asio, libdatachannel (WebRTC stack), Nix, Ansible
- Pet project (in development): configurable HFT matching engine focused on high-throughput, low-latency order matching for exchanges and arbitrage systems.
Stack: C++23, SIMD intrinsics, Cap'n Proto, Go

Core Software Engineer

Haarlem, Netherlands

AIMMS B.V.

Nov 2019 – Sep 2021

As a member of the AIMMS PRO core team:

1. AWS-to-Azure platform migration

- Configured RBAC for the Azure landing zone, bootstrapped Terraform state, and implemented supporting IaC.
- Migrated Java and Scala backend services from ECS to AKS, adapting them to the Kubernetes API.

2. Core C++ platform and developer tooling

- Built a C++17 microservice template with an API-first protobuf DSL used for both gRPC and REST surfaces.
- Worked on internal developer-platform infrastructure and package registries (Conan, Artifactory).
- Built PoCs and delivered demos to promote adoption of the developer-platform tooling.

Achievements:

- Improved cost efficiency and throughput predictability for compute-heavy optimization workloads with nondeterministic convergence by implementing Kubernetes-native bin packing through resource requests/limits and affinity rules, priority classes, and pre-warmed spare capacity using Cluster Autoscaler and taints/tolerations.
- Built migration tooling for sensitive, business-critical customer data in Amazon RDS.
- Built a declarative-schema CRUD generator for administrative and internal tools, reducing repetitive implementation work.

Stack: ECS, EC2, RDS, CloudWatch, Go, Python, Java, Scala, C++17, TypeScript, React, Conan, Artifactory, CMake, Ninja, gRPC-Gateway, Kubernetes (AKS), Application Insights, Terraform, Helm

Senior Software Engineer

Amsterdam, Netherlands

HVR SOFTWARE (ACQUIRED BY FIVETRAN)

Jan 2018 – Oct 2019

Implemented features and bug fixes in HVR's high-throughput, log-based data-replication engine.

- Contributed to core replication workflows.
- Reverse-engineered Oracle histogram metadata to support custom sharding functionality.
- Built a Go REST service for cloud integration.

Achievements:

- Rewrote a core feature-flag transpiler used in HVR's CI/CD pipeline, reducing runtime from over one minute to five seconds on a critical build path.

Stack: C99, Python, Go, RDBMS and big-data integrations, Oracle (histogram internals), Airflow, Spark

Software Engineer

Eindhoven, Netherlands

MAPSCAPE B.V.

2017

Implemented backend components of the Mapscape NDS compiler for raw map data parsing and introduced parallel parsing to reduce processing time.

Stack: C++11/14, Boost, GTest, SQLite, shell, Python, Airflow

Senior Software Engineer

St. Petersburg, Russia

NEW CLOUD TECHNOLOGIES

2014 – 2017

Built collaboration-engine features for an on-premises office suite and developed its spreadsheet client for Samsung Tizen.

Achievements:

- Built a cross-platform FreeType-based font-rendering engine with subpixel positioning for scalable document layouts and integrated it into a large enterprise codebase.
- Developed advanced spreadsheet editing for Samsung Tizen using a declarative, compile-time state machine to avoid dynamic dispatch overhead and resolve editing edge cases observed in competing web suites.

Stack: C++14, CMake, Emscripten, Objective-C, Qt, SWIG, Kubernetes

Senior Software Engineer

St. Petersburg, Russia

AIS

2014

Built C++11 microservices for the banking and exchange subsystem of a cryptocurrency mining pool:

- Designed an Asio-based service framework.
- Implemented multi-stage, fault-tolerant transactions.
- Reduced the risk of direct financial loss through append-only storage, two-factor transactions, circuit breakers, alerting, rollback paths, and extensive testing.
- Shipped the zero-to-one production release with a three-engineer team, owning the C++ subsystem end to end.

Achievements:

- Designed and implemented all C++11 services for the banking subsystem, including multi-stage transactions and circuit breakers with rollback paths.
- Delivered the production launch in under two months with end-to-end ownership of reliability under financial risk.

Stack: C++11, Core Java, Groovy, Ruby (Chef)

Education

B.Sc. in Computer Science and Engineering

Novosibirsk, Russia

NOVOSIBIRSK STATE TECHNICAL UNIVERSITY (NSTU)

2002 – 2007

Previous work history

AVAILABLE UPON REQUEST

2005 – 2014